# Chapter 8 Techniques of Integration

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## 8.1 Using Basic Integration Formulas

### Table 8.1: Basic Integration formulas

1. $\int k\,dx = kx + C$ (any number $k$)
2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$
3. $\int \frac{dx}{x} = \ln|x| + C$
4. $\int e^x\,dx = e^x + C$
5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \quad (a > 0, a \neq 1)$
6. $\int \sin x\,dx = -\cos x + C$
7. $\int \cos x\,dx = \sin x + C$
8. $\int \sec^2 x\,dx = \tan x + C$
9. $\int \csc^2 x\,dx = -\cot x + C$
10. $\int \sec x \tan x\,dx = \sec x + C$
11. $\int \csc x \cot x\,dx = -\csc x + C$
12. $\int \tan x\,dx = \ln\left|\sec x\right| + C$
13. $\int \cot x\,dx = \ln\left|\sin x\right| + C$
14. $\int \sec x\,dx = \ln\left|\sec x + \tan x\right| + C$
15. $\int \csc x\,dx = -\ln\left|\csc x + \cot x\right| + C$
16. $\int \sinh x\,dx = \cosh x + C$
17. $\int \cosh x\,dx = \sinh x + C$

**Table 8.1: Basic Integration formulas (Continue)**

**Integrals leading to inverse trigonometric functions or inverse hyperbolic functions**

The following formulas hold for any constant $a > 0$.

18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\left(\frac{x}{a}\right) + C \quad \left(\text{Valid for } x^2 < a^2\right)$

19. $\int \frac{dx}{a^2+x^2} = \frac{1}{a}\arctan\left(\frac{x}{a}\right) + C \quad$ (Valid for all $x$)

20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\operatorname{arcsec}\left|\frac{x}{a}\right| + C \quad (\text{Valid for } |x| > a > 0)$

21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C, \quad a > 0$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C, \quad x > a > 0$

23. $$\int \frac{dx}{a^2 - x^2} = \begin{cases} \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C, & x^2 < a^2 \\ \frac{1}{a}\coth^{-1}\left(\frac{x}{a}\right) + C, & x^2 > a^2 \end{cases}$$

24. $\int \frac{dx}{x\sqrt{a^2-x^2}} = -\frac{1}{a}\operatorname{sech}^{-1}\left(\frac{x}{a}\right) + C, \quad 0 < x < a$

25. $\int \frac{dx}{x\sqrt{a^2 + x^2}} = -\frac{1}{a}\operatorname{csch}^{-1}\left|\frac{x}{a}\right| + C, \quad x \neq 0 \text{ and } a > 0$

**Example 1 (Using Basic Integration Formulas and Basic Substitutions)**

(a) $\int_{-\sqrt{3}}^{\sqrt{2}} \frac{dx}{\sqrt{4-x^2}} = \arcsin\left(\frac{x}{2}\right)\Big]_{-\sqrt{3}}^{\sqrt{2}} = \arcsin\left(\frac{\sqrt{2}}{2}\right) - \arcsin\left(\frac{-\sqrt{3}}{2}\right) = \frac{\pi}{4} - \left(-\frac{\pi}{3}\right) = \frac{7\pi}{12}.$

(b) $\int \frac{dx}{\sqrt{5-4x^2}} = \int \frac{dx}{\sqrt{(\sqrt{5})^2-(2x)^2}} = \int \frac{\frac{du}{2}}{\sqrt{(\sqrt{5})^2-u^2}}$ (Let $u = 2x$. Then $du = 2dx$.)

$= \frac{1}{2}\arcsin\left(\frac{u}{\sqrt{5}}\right) + C = \frac{1}{2}\arcsin\left(\frac{2x}{\sqrt{5}}\right) + C.$

(c) $\int_0^1 \frac{2dx}{\sqrt{3+4x^2}} = \int_0^1 \frac{2dx}{\sqrt{(\sqrt{3})^2+(2x)^2}} = \int_0^2 \frac{2\left(\frac{du}{2}\right)}{\sqrt{(\sqrt{3})^2+u^2}}$ (Let $u = 2x$. Then $du = 2dx$. When $x = 0$, $u = 0$; when $x = 1$, $u = 2$.)

$= \sinh^{-1}\left(\frac{u}{\sqrt{3}}\right)\Big]_0^2 = \sinh^{-1}\left(\frac{2}{\sqrt{3}}\right) - \sinh^{-1}(0) \approx 0.98665.$

(d) $\int \frac{dx}{\sqrt{e^{2x}-6}} = \int \frac{dx}{\sqrt{(e^x)^2-(\sqrt{6})^2}} = \int \frac{du}{u\sqrt{u^2-(\sqrt{6})^2}}$ (Let $u = e^x$. Then $du = e^x dx \Rightarrow dx = \frac{du}{u}$.)

$= \frac{1}{\sqrt{6}}\text{arcsec}\left|\frac{u}{\sqrt{6}}\right| + C = \frac{1}{\sqrt{6}}\text{arcsec}\left(\frac{e^x}{\sqrt{6}}\right) + C.$

(e) $\int \coth 5x\, dx = \int \frac{\cosh 5x}{\sinh 5x} dx = \int \frac{du}{5u}$ (Let $u = \sinh 5x$. Then $du = 5\cosh x dx$.)

$= \frac{1}{5}\ln|u| + C = \frac{1}{5}\ln|\sinh 5x| + C.$

(f) $\int_3^5 \frac{2x-3}{\sqrt{x^2-3x+1}} dx = \int_1^{11} \frac{du}{\sqrt{u}} = \int_0^{11} u^{-1/2} du$ (Let $u = x^2 - 3x + 1$. Then $du = (2x - 3)dx$. When $x = 3$, $u = 1$; when $x = 5$, $u = 11$.)

$= 2\sqrt{u}\Big]_1^{11} = 2\left(\sqrt{11} - 1\right).$

**Example 2 (Completing the Squares)**

Evaluate (a) $\int \frac{dx}{\sqrt{4x-x^2}}$ (b) $\int \frac{dx}{4x^2+4x+2}$

Solution:

Both integrals in (a) and (b) do not match any of the formulas in Table 8.1, so we first need to rewrite the quadratic expressions by completing the squares.

(a) $\int \frac{dx}{\sqrt{4x-x^2}} = \int \frac{dx}{\sqrt{-(x^2-4x)}} = \int \frac{dx}{\sqrt{-[(x-2)^2-(2)^2]}} = \int \frac{dx}{\sqrt{(2)^2-(x-2)^2}}$ (Let $u = x - 2$. Then $du = dx$.)

$= \int \frac{du}{\sqrt{(2)^2-u^2}} = \arcsin\left(\frac{u}{2}\right) + C = \arcsin\left(\frac{x-2}{2}\right) + C.$

(b) $\int \frac{dx}{4x^2+4x+2} = \int \frac{dx}{4(x^2+x+\frac{1}{2})} = \int \frac{dx}{4\left[\left(x+\frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2 + \frac{1}{2}\right]}$

$= \int \frac{dx}{4\left(x+\frac{1}{2}\right)^2 + 1} = \int \frac{dx}{(2x+1)^2+1}$ (Let $u = 2x + 1$. Then $du = 2dx$.)

$= \int \frac{du/2}{u^2+1} = \frac{1}{2}\arctan u + C = \frac{1}{2}\arctan(2x+1) + C.$

**Example 3 (Use Identities)**

(a) $\int_0^1 \sinh^2 x\,dx = \int_0^1 \frac{\cosh 2x-1}{2}dx = \frac{1}{2}\left[\frac{\sinh 2x}{2} - x\right]_0^1 = \frac{1}{2}\left(\frac{\sinh 2}{2} - 1\right) \approx 0.40672.$

(b) $\int(\cos x \sin 2x + \sin x \cos 2x)dx = \int \sin(x+2x)\,dx = \int \sin 3x\,dx = -\frac{\cos 3x}{3} + C.$

**Example 4 (Multiply by a Form of 1)**

Evaluate $\int_0^{\pi/4} \frac{dx}{1-\sin x}dx.$

Solution:
Multiply the numerator and denominator of the integrand by 1 + sin $x$ to transform the integral into one we can evaluate:

$$\int_0^{\pi/4} \frac{dx}{1-\sin x} = \int_0^{\pi/4} \frac{1}{1-\sin x}\cdot\frac{1+\sin x}{1+\sin x}dx$$ Multiply and divide by conjugate.

$$= \int_0^{\pi/4} \frac{1+\sin x}{1-\sin^2 x}dx$$ Simplify.

$$= \int_0^{\pi/4} \frac{1+\sin x}{\cos^2 x}dx$$ $1 - \sin^2 x = \cos^2 x$

$$= \int_0^{\pi/4} \left(\sec^2 x + \sec x \tan x\right)dx$$ Use Table 8.1, Formulas 8 and 10

$$= [\tan x + \sec x]_0^{\pi/4} = (1 + \sqrt{2} - (0+1)) = \sqrt{2}.$$

**Example 5 (Improper Fractions)**

Evaluate $\int \frac{3x^2-7x}{3x+2}dx.$

Solution:
The integrand is an improper fraction since the degree of the numerator is greater than the degree of the denominator. To integrate it, we can perform a long division to obtain a quotient plus a remainder that is a proper fraction:

$$\int \frac{3x^2-7x}{3x+2}\,dx = \int (x - 3 + \frac{6}{3x+2})\,dx = \frac{x^2}{2} - 3x + 2\ln\left|3x+2\right| + C.$$

$$\begin{array}{r} x-3 \\ 3x+2\overline{)\,3x^2-7x} \\ \underline{3x^2+2x} \\ -9x \\ \underline{-9x-6} \\ +6 \end{array}$$

**Remark:**

Reducing an improper fraction by long division does not always lead to an expression we can integrate directly. We see what to do about that in Section 8.5.

**Example 6 (Separate the Integrand)**

Evaluate $\int \frac{3x+2}{\sqrt{1-x^2}}dx$.

Solution:
Separate the integrand to get

$$\int \frac{3x+2}{\sqrt{1-x^2}}dx = \int \frac{3x}{\sqrt{1-x^2}}dx + \int \frac{2}{\sqrt{1-x^2}}dx$$ (Let $u = 1 - x^2$. Then $du = -2xdx$.)

$$= 3\int \frac{-du/2}{\sqrt{u}}du + 2\arcsin x + C = -\frac{3}{2}\int u^{-\frac{1}{2}}du + 2\arcsin x + C$$

$$= -3u^{1/2} + 2\arcsin x + C = = -3\sqrt{1-x^2} + 2\arcsin x + C.$$

The question of what to substitute for in an integrand is not always quite so clear. Sometimes we may need to use another method after a substitution.

**Example 7**

Evaluate $\int \frac{dx}{(1+\sqrt{x})^3}$.

Solution:
We might try substituting for the term $\sqrt{x}$, but the derivative factor $1/\sqrt{x}$ is missing from the integrand, so this substitution will not help. The other possibility is to substitute for (1+ $\sqrt{x}$,), and it turns out this works:

$$\int \frac{dx}{(1+\sqrt{x})^3} = \int \frac{2(u-1)}{u^3}du$$ (Let $u = 1 + \sqrt{x}$. Then $du = -\frac{dx}{2\sqrt{x}} \Rightarrow dx = -2(1-u)du$ .)

$$= 2\int \left(\frac{1}{u^2} - \frac{1}{u^3}\right)du = -\frac{2}{u} + \frac{1}{u^2} + C = \frac{-2u+1}{u^2} + C = \frac{1-2\left(1+\sqrt{x}\right)}{\left(1+\sqrt{x}\right)^2} + C$$

**Example 8**

$$\int_0^{\ln 2} 4e^x \sinh x\, dx = \int_0^{\ln 2} 4e^x \frac{e^x - e^{-x}}{2}dx = \int_0^{\ln 2} (2e^{2x} - 2)\, dx$$

$$= \left[e^{2x} - 2x\right]_0^{\ln 2} = (e^{2\ln 2} - 2\ln 2) - (1 - 0)$$

$$= 4 - 2\ln 2 - 1 \approx 1.6137$$

**Example 9**

Evaluate $\int_{-\pi/2}^{\pi/2} x^3 \cos x dx$.

Solution:
No substitution or algebraic manipulation is clearly helpful here. But we observe that the interval of integration is the symmetric interval [−π/2, π/2] and the integrand is an odd function.

Let $f(x) = x^3 \cos x$. Since $f(-x) = (-x)^3 \cos(-x) = -x^3 \cos x = -f(x)$ for all $x \in [-\pi/2, \pi/2]$, then, $\int_{-\pi/2}^{\pi/2} x^3 \cos x dx = 0$ by Theorem in Chapter 5.

**Example 10 (Assorted Integrations)**

Evaluate

(1) $\int_0^{1/4} e^{\tan \pi\theta} \sec^2 \pi\theta d\theta$

(2) $\int_{\sqrt{2}}^{2} \frac{\sec^2(\sec^{-1} x)dx}{x\sqrt{x^2-1}}$

(3) $\int \frac{dx}{2\sqrt{x}+2x}$

(4) $\int_0^2 \sinh^2 2x dx$

(5) $\int_0^{\ln 4} e^x \cosh x dx$

Solution:

(1) Let $u = \tan \pi\theta$. Then $du = \pi \sec^2 \pi\theta d\theta$. When $\theta = 0$, $u = 0$; when $\theta = 1/4$, $u = 1$.

$$\int_0^{1/4} e^{\tan \pi\theta} \sec^2 \pi\theta d\theta = \int_0^1 e^u \frac{du}{\pi} = \frac{e^u}{\pi}\Big]_0^1 = \frac{1}{\pi}\left(e^1 - e^0\right) = \frac{1}{\pi}(e-1).$$

(2) Let $u = \sec^{-1} x$. Then $du = \frac{dx}{x\sqrt{x^2-1}}$.

When $x = \sqrt{2}$, $u = \sec^{-1}\sqrt{2} = \cos^{-1}(1/\sqrt{2}) = \pi/4$;

When $x = 2$, $u = \sec^{-1} 2 = \cos^{-1}(1/2) = \pi/3$.

$$\int_{\sqrt{2}}^{2} \frac{\sec^2(\sec^{-1} x)dx}{x\sqrt{x^2-1}} = \int_{\pi/4}^{\pi/3} \sec^2 u\, du = \tan x\Big]_{\pi/4}^{\pi/3} = \tan\frac{\pi}{3} - \tan\frac{\pi}{4} = \sqrt{3}-1 \ .$$

(3) $\int \frac{dx}{2\sqrt{x}+2x} = \int \frac{dx}{2\sqrt{x}\left(1+\sqrt{x}\right)} = \int \frac{du}{u}$ where $u = 1+\sqrt{x}$ and $du = \frac{1}{2\sqrt{x}}dx$

$$= \ln|u| + C = \ln|1+\sqrt{x}| + C = \ln(1+\sqrt{x}) + C \qquad (\because\ 1+\sqrt{x} > 0 \ \ \forall x > 0)$$

(4) $\int_0^2 \sinh^2 2x dx = \int_0^2 \frac{\cosh 4x - 1}{2} dx = \frac{1}{2}\left[\frac{\sinh 4x}{4} - x\right]_0^2 = \frac{1}{2}\left[\frac{\sinh 8}{4} - 2\right] \approx 185.3.$

(5) $\int_0^{\ln 4} e^x \cosh x dx = \int_0^{\ln 4} \frac{e^x(e^x + e^{-x})}{2} dx = \frac{1}{2}\int_0^{\ln 4} (e^{2x}+1)dx = \frac{1}{2}\left[\frac{e^{2x}}{2} + x\right]_0^{\ln 4}$

$$= \frac{1}{2}\left[\frac{e^{2\ln 4}}{2} + \ln 4 - \frac{1}{2}\right] = \frac{1}{2}\left[\frac{e^{\ln 16}}{2} + 2\ln 2 - \frac{1}{2}\right] = \frac{15}{4} + \ln 2.$$